

Hardware CWE SIG Meeting Notes

Meeting Minutes

September 18, 2026

12:30pm – 1:30pm EST

Meeting Attendance

Irena Bojanova	Abraham Fernandez	Parbati K. Manna
Steven Christey Coley	Rubio	Bruce Monroe
Amitabh Das	Alexander Harner	Jeremy Muldavin
Daniel DiMase	Bob Heinemann	Martin Thompson
	Rachana Maitra	

Agenda

- Housekeeping and CWE 5.0 updates
- Simulation Synthesis Weakness Update
- Hardware Lifecycle Phase Definitions

Meeting Notes

Housekeeping and CWE 5.0 updates

Schedule and release planning

Bob announced that the next Hardware CWE SIG meeting is October 15 and said the CWE 5.0 release remains planned for the October time frame. A specific release date had not yet been set.

Content repository and staged hardware updates

Bob reminded participants that the public CWE Content Development Repository is used to process submissions and corrections and can be viewed and commented on. He also said the Hardware View

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flattening is staged in the development environment for CWE 5.0, along with updates to PeerOf and CanAlsoBe relationships among related hardware CWEs.

Simulation Synthesis Weakness Update

Review progress

Bob reported substantial reviews, edits, and updates to the simulation-synthesis weakness entry. He said most elements now contain high-quality data and that the working group is focused primarily on demonstrative examples, code-snippet presentation, attribution language, and other stylistic comments. He said the entry is planned for CWE 5.0.

RTL and HDL terminology

Martin reiterated his concern that the term RTL is narrower than the intended scope and asked whether HDL could be used instead. Bob said many entry updates had already been changing RTL-oriented wording to HDL, although that terminology change had not yet been propagated to the slide title.

Hardware Lifecycle Phase Definitions

Purpose and use of phases in CWE

Bob framed CWE lifecycle phases as metadata used to communicate where a weakness is introduced, where it can be mitigated, and, beginning with CWE 5.0, where it can be detected. He emphasized that the phase values are independent labels rather than an ordered development-process model. The purpose of the hardware-specific refinement is to improve how CWE communicates root cause, mitigation opportunities, detection activities, and assessment-relevant artifacts.

Feedback on the initial hardware draft

Bob summarized prior feedback that the initial draft was too silicon-specific. He said the terminology needed to cover whole-device hardware more effectively, account for component selection as a possible source of weaknesses and avoid conflating pre-silicon IP or SoC integration with later device or system integration. He had incorporated comments into a revised draft circulated before the meeting.

Hardware Reuse and Decommissioning

The discussion supported treating Hardware Reuse and Decommissioning as separate phases in the current draft. Daniel emphasized that a reuse definition should account for legitimate and illegitimate reuse and raised the case of reused parts represented as new. Bob said counterfeit-related detail may be important to CWE but would require additional thought about whether it belongs in a phase definition, a technical note, or weakness-specific content. Steve explained that phase definitions are expected to appear as schema documentation strings for CWE 5.0, accessible through schema documentation rather than directly inside each CWE entry.

Hardware Requirements

Bob proposed shortening the phase name Hardware Requirements Definition to Hardware Requirements. No objections were raised, and he treated the shorter name as the working choice for the draft.

Architecture and microarchitecture

Abraham argued for separating hardware architecture from hardware microarchitecture because weaknesses can be introduced independently at either level. He described architecture as what a processor or product does and microarchitecture as how it implements that behavior. Steve suggested using the CWE purpose as a test: if introduction, mitigation, or detection can occur in one phase but not the other, that supports separate phase labels. Bob concluded that, for the next iteration, Hardware Architecture and Hardware Microarchitecture should remain separate.

Component and IP selection

Participants debated where component and IP selection belong. Martin noted that, from a board-design perspective, component selection can mean selecting items such as microcontrollers, memory, accelerators, FPGAs, or SoCs, while Abraham used component selection in a microarchitectural context. Rachana favored keeping phase names simple and expressing detailed activities in definitions. The discussion moved away from creating standalone Component Selection or IP Selection phases; Bob tentatively moved component selection into the microarchitecture discussion, while participants recognized that the terms remain overloaded and need further refinement.

Chip-design and board-design perspectives

Martin cautioned that microarchitecture is familiar in processor and chip design but is not a normal board-design term. He suggested that users designing boards may need terminology that lets them find weaknesses relevant to activities such as board architecture and schematic capture. Bob said the draft had originated from a microelectronics-oriented perspective and acknowledged that board-design concerns need to be incorporated. He asked whether there is a publicly available board-design lifecycle that could be used as a reference, and Martin agreed to look for one.

Terminology and granularity

The group discussed the risk of both excessive granularity and overly broad labels. Martin suggested high-level design and low-level design as possible alternatives, while Steve cautioned that those terms may be interpreted differently across communities. Bob noted that a CWE entry can carry multiple phase values, which may reduce the risk of forcing a weakness into a single overly precise phase. Participants also emphasized that the value of added precision should be judged by how CWE users will apply the phase information.

Discussion outcome

Bob closed by saying the comments should be taken offline for another iteration. He plans to consider chip-design and board-design perspectives separately and then determine how to merge them into a hardware vocabulary that works across communities. He identified a distinct microarchitecture phase

as an important working direction, while noting that the remaining phase definitions and activity placement still need refinement. No final hardware lifecycle vocabulary was approved at this meeting.

Decisions Made

1. Use Hardware Requirements as the working phase name instead of Hardware Requirements Definition.
2. Keep Hardware Architecture and Hardware Microarchitecture as separate phase concepts in the next proposal iteration.
3. Keep Hardware Reuse and Decommissioning separated in the working draft; further refine how reuse, counterfeit-related concerns, and supporting definitions are documented.
4. Do not add standalone Component Selection or IP Selection phases currently. Continue refining how those activities are represented within the broader phase definitions.
5. Continue iterating on the hardware lifecycle proposal rather than approving a final phase vocabulary at this meeting.

Action Items:

- Revise the hardware lifecycle phase proposal to incorporate the meeting feedback, including the chip-designer and board-designer perspectives, and circulate a new iteration for review. (Bob)
- Look for and share a publicly available board-design lifecycle or comparable reference that can inform the phase vocabulary. (Martin)

Next Steps

Continue the phase-definition work over email before the next SIG meeting, with particular attention to terminology that is meaningful across both chip and board design communities.

Refine the placement and definitions of component selection, IP selection, architecture, microarchitecture, board-oriented activities, reuse, and decommissioning without turning CWE phase metadata into a prescriptive development-process model.

The simulation-synthesis weakness remains planned for CWE 5.0, along with the staged Hardware View flattening and relationship updates. The next Hardware CWE SIG meeting is October 15, 2026.